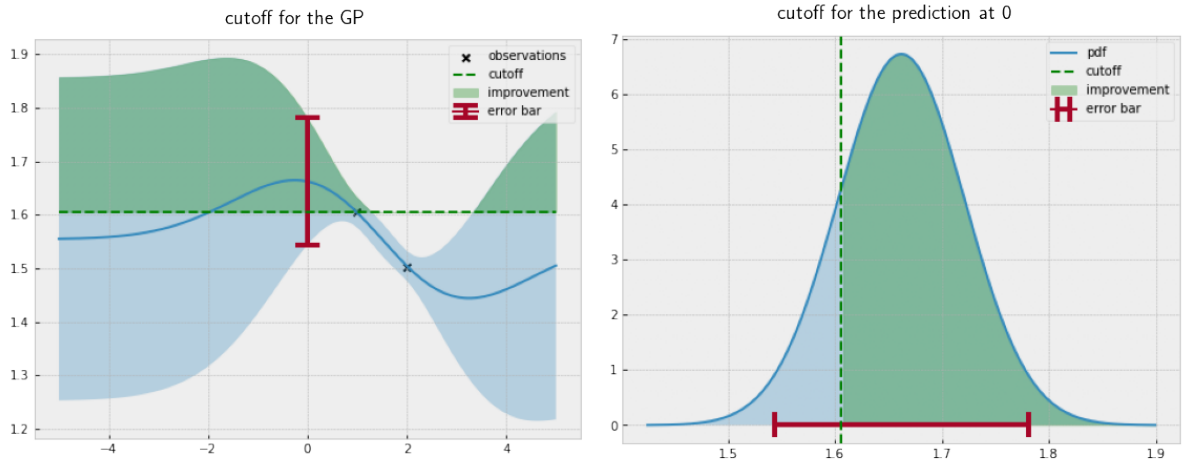


Figure 4.9 Improvement from the incumbent at 0, highlighted in a darker shade. The left panel shows the entire Gaussian process, while the right panel only shows the normal distribution corresponding to the prediction at 0 (the error bars are the same across the two panels). Here, improvement from the incumbent follows a truncated normal distribution.

These conveniences don't only apply to $x = 0$. Since the prediction made by the GP at any given point is a normal distribution, the improvement at any point also follows a normal distribution that is cut off, as shown in Figure 4.10.

We see that compared with Figure 4.9, where a large portion of the normal distribution at 0 is highlighted as possible improvements:

- The prediction at -2 (center panel) is worse in the sense that only half of it shows possible improvements. This is because the mean prediction at -2 is roughly equal to the incumbent, so there's only a 50-50 chance that the function value at -2 will improve from 1.6.
- As another example, the right panel shows the prediction at 3, where it is almost impossible to improve from the incumbent, as almost the entire normal distribution at 3 lies below the incumbent threshold.

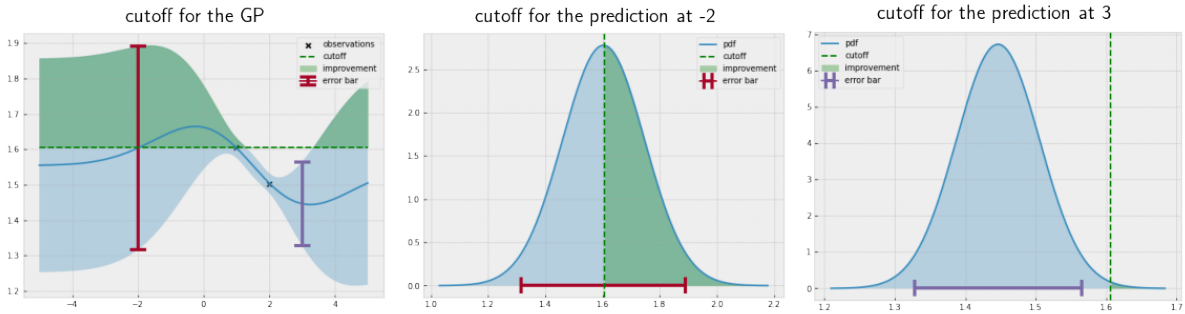


Figure 4.10 Improvement from the incumbent at -2 and 3, highlighted in a darker shade. The left panel shows the entire Gaussian process, the center panel shows the prediction at -2, and the right panel shows the prediction at 3. The highlighted portions show possible improvements, which depend on the normal distribution at a given point.

This shows that different points will lead to different possible improvements depending on the

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